# Operating System Lab – 05

## Topic: Loop, Sum, Even/Odd, Factorial
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**Youtube Channel: https://www.youtube.com/@CodionTeam**

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## 1 While Loop (Bash)

### 🔹 Bash Code

```
n=1
while [ $n -le 6 ]
do
echo "$n"
n=$((n+1))
done
```

### 🔹 Explanation

- `n=1` দিয়ে loop শুরু হয়েছে
- যতক্ষণ `n ≤ 6`, loop চলবে
- প্রতি iteration এ `n` print হবে
- `n=$((n+1))` দিয়ে মান বাড়ানো হচ্ছে

👉 Output: `1 2 3 4 5 6`

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### 🔹 Same Logic in C Language

```
#include <stdio.h>

int main() {
    int n = 1;
    while (n <= 6) {
        printf("%d\n", n);
        n++;
    }
    return 0;
```

```
}
```

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# 2 For Loop (Range Based)

### ◆ Bash Code

```bash
for i in {1..10..2}
do
echo "$i"
done
```

### ◆ Explanation

- `{1..10..2}` → 1 থেকে 10 পর্যন্ত, প্রতি ধাপে 2 করে বাড়ে
- Output হবে: `1 3 5 7 9`

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### ◆ Same Logic in C

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 10; i += 2) {
        printf("%d\n", i);
    }
    return 0;
}
```

---

# 3 C-Style For Loop in Bash

### ◆ Bash Code

```bash
for ((i=0; i<=5; i++))
do
echo "$i"
done
```

### ◆ Explanation

- এটি একদম C language এর মতো loop
- 0 থেকে 5 পর্যন্ত সংখ্যা print হবে

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### ◆ Same Logic in C

```
#include <stdio.h>

int main() {
    for (int i = 0; i <= 5; i++) {
        printf("%d\n", i);
    }
    return 0;
}
```

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## 4 Sum of Numbers (0–50) & Even/Odd Check

### ◆ Bash Code

```
sum=0
for ((i=0; i<=50; i++))
do
sum=$((sum+i))
done
echo "Addition is: $sum"

if ((sum % 2 == 0))
then
echo "This is even number"
else
echo "This is odd number"
fi
```

### ◆ Explanation

- `sum=0` দিয়ে যোগ শুরু
- 0 থেকে 50 পর্যন্ত সব সংখ্যা যোগ করা হয়েছে
- `%` দিয়ে Even / Odd চেক

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### ◆ Same Logic in C

```
#include <stdio.h>

int main() {
    int sum = 0;
    for (int i = 0; i <= 50; i++) {
        sum += i;
    }

    printf("Addition is: %d\n", sum);
```

```
    if (sum % 2 == 0)
        printf("This is even number\n");
    else
        printf("This is odd number\n");

    return 0;
}
```

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# 5 User Input Sum & Even/Odd

## ◆ Bash Code

```bash
sum=0
for ((i=1; i<=5; i++))
do
read -p "Enter number $i: " num
sum=$((sum+num))
done

echo "Addition is: $sum"

if ((sum % 2 == 0))
then
echo "This is even number"
else
echo "This is odd number"
fi
```

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## ◆ Same Logic in C

```c
#include <stdio.h>

int main() {
    int sum = 0, num;

    for (int i = 1; i <= 5; i++) {
        printf("Enter number %d: ", i);
        scanf("%d", &num);
        sum += num;
    }

    printf("Addition is: %d\n", sum);

    if (sum % 2 == 0)
        printf("This is even number\n");
    else
        printf("This is odd number\n");

    return 0;
}
```

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# 6 Print Even Numbers from 1 to N

### ◆ Bash Code (Corrected)

```
read -p "Enter the range: " N
for ((i=1; i<=N; i++))
do
if ((i % 2 == 0))
then
echo "$i"
fi
done
```

### ◆ Explanation

- User range input নেয়
- `% 2 == 0` হলে even number print

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### ◆ Same Logic in C

```
#include <stdio.h>

int main() {
    int N;
    printf("Enter the range: ");
    scanf("%d", &N);

    for (int i = 1; i <= N; i++) {
        if (i % 2 == 0)
            printf("%d\n", i);
    }
    return 0;
}
```

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# 7 Factorial Using Loop

**Mathematical Logic:** n! = n * (n-1) * ……..*1

### ◆ Bash Code (Corrected)

```
read -p "Enter a number: " num
fact=1

for ((i=1; i<=num; i++))
do
```

```
fact=$((fact*i))
done

echo "Factorial of $num = $fact"
```

---

## ◆ Same Logic in C

```c
#include <stdio.h>

int main() {
    int num;
    long long fact = 1;

    printf("Enter a number: ");
    scanf("%d", &num);

    for (int i = 1; i <= num; i++) {
        fact *= i;
    }

    printf("Factorial of %d = %lld\n", num, fact);
    return 0;
}
```

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# 🔁 Loop & Conditional Statements: C vs Bash

## 1️⃣ for Loop

| Feature | C Language | Shell Script (Bash) |
|---|---|---|
| **Syntax** | `for(init; condition; inc)` | `for ((init; condition; inc))` |
| **Example** | `for(i=0; i<5; i++)` | `for ((i=0; i<5; i++))` |
| **Block Start** | `{` (Curly Brace) | `do` |
| **Block End** | `}` | `done` |
| **Semicolon** | Mandatory | Not used |
| **Data Type** | Required (e.g., `int`) | Not required |

## 2️⃣ while Loop

| Feature | C Language | Shell Script (Bash) |
|---|---|---|
| **Syntax** | `while(condition)` | `while [ condition ]` |
| **Block Start** | `{` | `do` |
| **Block End** | `}` | `done` |

## 3 if Statement

| Feature | C Language | Shell Script (Bash) |
|---|---|---|
| **Syntax** | `if(condition)` | `if [ condition ]` |
| **Condition Symbols** | `<`, `>`, `==` | `-lt`, `-gt`, `-eq` |
| **Block Start** | `{` | `then` |
| **Block End** | `}` | `fi` |
| **Semicolon** | Required | Not required |

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## 4 if–else & if–else–if Statement

| Feature | C Language | Shell Script (Bash) |
|---|---|---|
| **Else-if Keyword** | `else if` | `elif` |
| **Else Keyword** | `else` | `else` |
| **Block Ending** | `}` | `fi` |